



CAREER PREDICTION USING MACHINE LEARNING

Internship Project Report

1. Project Title:

Career Prediction Using Machine Learning Classification Models

2. Objective:

The objective of this project is to develop a machine learning model that predicts a student's suitable career path based on their interests, skills, coding level, and academic background. The system aims to assist in early career guidance using data-driven predictions.

3. Problem Statement:

Students often face difficulty in choosing the right career path due to lack of guidance and clarity about their strengths. This project addresses this issue by building a predictive model that analyses key attributes and suggests an appropriate career domain.

4. Dataset Description:

A structured dataset was created containing the following features:

Input Features:

- Interest (e.g., AI, Web Development, Data Science)
- Skill (e.g., Python, Machine Learning, HTML, JavaScript)
- Coding Level (Beginner, Intermediate, Advanced)
- Subject Strength (e.g., Mathematics, Computer Science)

Target Variable:

- Career (AI Engineer, Web Developer, Data Scientist)

5. Methodology

5.1 Data Preprocessing:

- Categorical data was converted into numerical format using Label Encoding.
- Dataset was split into training and testing sets.
- Features and target variables were separated.

5.2 Model Selection

Two machine learning models were used:

- Logistic Regression
- Random Forest Classifier

5.3 Training Process

- Models were trained using the training dataset
- Predictions were made on the test dataset

6. Evaluation Metrics

The performance of the models was evaluated using:

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix

7. Results

Logistic Regression	- accuracy(1.0%)
Random Forest Classifier	- accuracy(1.0%)

Best Performing Model: Random Forest Classifier (due to higher accuracy and better handling of feature relationships)

8. Visualization

The following visualizations were used:

- Class distribution of target variable
- Confusion matrix heatmap for model evaluation
- These graphs helped in understanding dataset balance and model performance.

10. Future Improvements

- Use larger and real-world datasets
- Add more career categories
- Deploy model as a web application using Flask or Streamlit
- Improve accuracy using hyperparameter tuning

11. Tools & Technologies Used

- Python
- Pandas, NumPy
- Scikit-learn
- Matplotlib, Seaborn
- Jupyter Notebook

Conclusion

The machine learning model successfully predicts career paths based on student attributes with reasonable accuracy. The Random Forest Classifier performed better compared to Logistic Regression due to its ability to handle complex feature interactions.

This system demonstrates the practical application of machine learning in career guidance and decision support systems.

Scikit-learn

Matplotlib, Seaborn

Jupyter Notebook

Author

Internship Project – Machine Learning Task 1